Token-Efficient Context Sharing in Multi-Agent LLM Systems: How Much Can Be Cut, and What Limits It

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Abstract

Multi-agent LLM systems coordinate by forwarding context between agents, and every forwarded token is billed. Common practice forwards whole conversation histories, because deciding what to withhold is hard, so coordination cost grows quadratically in the number of agents. We ask a practical question: how much of that traffic can be removed without paying for it in task quality, and what stops us removing more?

On a public multi-hop benchmark reframed as a multi-receiver allocation problem, we find that a 2.7× reduction in inter-agent tokens costs no measurable accuracy, and that an oracle allocation reaches 21× while *improving* accuracy — so over-delivery is not merely wasteful but mildly harmful, and roughly an order of magnitude is available in principle. Pushing a practical policy to 5.4× does cost 8 accuracy points, so the frontier is real and we characterise it.

How the budget is spent matters more than how much is spent. Allocating per receiver beats selecting one globally-scored set for everyone by 19.2 accuracy points at matched cost. We explain this with a separation result: the one-set formulation — the natural reading of the problem as influence maximization — loses a factor $\Theta(n)$ in the number of receivers, under two independent mechanisms, and does so even on instances where the objective is monotone and submodular. We also show the two standard optimisation toolchains are unavailable here, the second because its structural parameters do not exist rather than because its bound is loose.

Finally, a negative result that we think is the most useful finding for practitioners. Across four allocation algorithms, including a resource-allocation program that exploits structure no baseline can, none beats a tuned uniform per-receiver rule at matched budget; and replacing lexical relevance with a dense retriever makes matters worse. Three independent lines of evidence place the binding constraint on *estimating* which agent needs what, not on optimising the allocation. The remaining order of magnitude is a retrieval problem, and effort spent on cleverer allocation will not recover it.

Keywords

multi-agent systems; large language models; communication; resource allocation; submodular optimization

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1 Introduction

A multi-agent LLM system coordinates by passing context. A planner tells a solver what the constraints are; an auditor is handed the records it must check. Every such transmission is billed per token, so coordination is a direct operating cost, and the prevailing engineering answer — forward the whole conversation, because deciding what to withhold is hard and sending everything is safe — makes that cost grow quadratically in the number of agents.

This paper treats that as the problem to solve. We ask how much inter-agent traffic can be removed before task quality suffers, what the shape of the cost–quality frontier is, and what prevents us from reaching the cheap allocations that clearly exist.

What we find. Three things, in decreasing order of how comfortable they are.

First, **a lot of the traffic is free to remove**. On a public multi-hop benchmark reframed as multi-receiver allocation, cutting inter-agent tokens by 2.7× costs no measurable accuracy, and an oracle allocation — which knows which item each agent needs — reaches 21× while *improving* accuracy by about five points. Over-delivery is therefore mildly harmful and not merely expensive, which is consistent with the measured tendency of LLMs to degrade under irrelevant context [16, 17, 22]. An order of magnitude is available in principle.

Second, **how the budget is spent matters more than how much**. Allocating a possibly different set to each receiver beats choosing one globally-scored set for everyone by 19.2 accuracy points at matched cost. This is not a tuning detail: we prove the one-set formulation loses a factor $\Theta(n)$ in the number of receivers (Proposition 1), under two independent mechanisms, and even on instances where the objective is monotone and submodular. That formulation is the natural reading of context sharing as influence maximization [13], so a substantial and otherwise applicable literature [1, 6, 20] turns out to optimise the wrong object here. We also show that the nearest general framework for the messy objective this problem actually has is unavailable, because its structural parameters fail to exist rather than merely giving a loose bound (§5).

Third, and most useful to anyone building such a system, **the remaining order of magnitude is not an optimisation problem**. We implement four allocation algorithms, including one that exploits separability structure no per-receiver baseline can imitate, and none beats a tuned uniform per-receiver rule at matched budget. Replacing lexical relevance with a dense retriever makes per-receiver delivery *worse*. Three independent observations converge: the binding constraint is estimating which agent needs what.

The oracle fixes the scale — full delivery on 208 tokens where the best implementable policy reaches 62% on 929 — and no allocation algorithm closes that gap, because the information required is absent from the scores rather than badly exploited.

Contributions. A measured cost–quality frontier for inter-agent context sharing, including the free reduction and the oracle ceiling (§7); a separation result showing the one-set formulation is off by $\Theta(n)$ and two degeneracy results removing the standard toolchains (§4, §5); a decomposition showing the per-receiver problem factors exactly through a scalar budget price (§6); and a negative result locating the residual gap in relevance estimation rather than allocation (§7).

2 Related work

Influence maximization. Monotonicity and submodularity of the spread function are what license greedy's $(1-1/e)$ [13], and the scaling literature [1, 6, 20] inherits those assumptions. Recent work relaxes them in various directions but keeps the seed-set object: one chosen set, serving the whole network. Our separation is orthogonal to the quality of those relaxations.

Non-monotone and non-submodular maximization. Guarantees exist for non-monotone submodular maximization [2, 8], for weakly submodular objectives via the submodularity ratio [4, 5] or supermodular degree [7], and for differences of submodular functions [11]. Closest to our setting, Shi and Lai [19] treat non-monotone, non-submodular maximization under a knapsack. §5 shows their parameters do not exist here. Regularized objectives of the form $g - c$ with g submodular and c modular admit distorted-greedy guarantees [10]; the density rule we discuss in §6 is an instance of that lineage, and we credit it as such rather than presenting it as new.

Context and communication cost in LLM agents. Empirical work cuts inter-agent tokens by pruning the message graph [23], generating the topology per task [12], or sharing key–value caches instead of text [18]; none states an optimality guarantee. Closest to our applied setting, RCR-Router selects a memory subset per agent under a strict token budget and reports up to 30% savings on multi-hop QA, again without guarantees [15]. A parallel line does prove guarantees, but for a *single* receiver: PACMS treats context assembly as submodular selection [9], and BPS maximizes a monotone submodular benefit minus a modular token penalty under a hard budget, with a bicriteria $(1-1/e, 1)$ ratio [3]. BPS is the single-receiver, modular-penalty, submodular-benefit special case of the problem we study, and it is the closest prior work; it observes that redundant context "can even degrade performance" and then charges that degradation as a separate cost, keeping the benefit submodular. Both halves of the decomposition are where our setting departs. Its penalty $\kappa_E\,\ell(S)$ is linear in total length, with κ_E an explicitly *first-order* per-token sensitivity, whereas the degradation we measure accelerates with load (§3); and its benefit is a weighted coverage $\sum_k w_k h_k(\sum_{i\in S} u_{i,k})$ with each h_k concave, so items overlap within a dimension and are *additive* across dimensions. Two items that are separately inert but jointly decisive have no representation in that form, which is the case §5 turns on. On the theory side, [14] treats finite context windows as hard fan-in limits and locates saturation via a mixing depth, but through majority-vote aggregation rather than set-function optimization. We are not aware of a formal treatment of the multi-receiver token-allocation problem, nor of an influence-maximization framing of LLM-agent communication.

3 Model

Receivers (agents) are indexed $1, \ldots, n$. A pool F of information items is given, item f costing $c(f) > 0$ tokens. An *allocation* assigns a set $S_i \subseteq F$ to each receiver. Information is copyable, but each transmission is charged, so total spend is $\sum_i c(S_i)$ and serving n receivers with one set costs n times serving one. We also consider a *free-broadcast* convention, where a set may be given to everyone for one transmission, to show our results do not rest on the cost model.

Per-receiver utility is

$$u_i(S) = \mathrm{rel}_i(S) - \rho_i(\mathrm{conf}_i(S)), \qquad (1)$$

relevance minus a saturation penalty. Two features of (1) are empirically motivated rather than assumed for convenience. First, ρ_i is increasing: delivering more context degrades an agent [17, 22]. Second, the penalty's argument is *confusable* load conf_i, not raw token count, because length-matched studies find that semantic proximity to the task predicts degradation while raw length barely does [16]. Cost and damage are therefore different quantities: one is billed and modular, the other is semantic and per-receiver. The objective is $U = \sum_i u_i(S_i)$ subject to the budget.

Example 1. Three agents share a pool of 20 documents. Agent 1 must check a delivery date, agent 2 a payment total, agent 3 must reconcile the two. Agents 1 and 2 each need one document; agent 3 needs both. A uniform allowance of one document per agent starves agent 3; an allowance of two overfeeds agents 1 and 2 with a document that can only distract them. No single set serves all three, and the cheapest correct allocation sends different sets to different agents. This is the structure Proposition 1 makes precise.

4 The seed-set formulation loses a factor n

Proposition 1. *For every $n \geq 1$ there is an instance with n receivers on which, at an identical budget,*

$$\frac{\mathrm{OPT}_{\textit{per-receiver}}}{\mathrm{OPT}_{\textit{seed-set}}} = n.$$

The separation holds under both cost conventions, for two independent reasons.

Proof. Take n receivers, n distinct items each of cost c, and let receiver i be satisfied only by item i. Set the budget $B = nc$.

(a) Billed per receiver. Per-receiver allocation sends item i to receiver i, spending $nc = B$ for utility n. A seed set S reaches all n receivers and so costs $n|S|c \leq nc$, forcing $|S| \leq 1$; a single item satisfies exactly one receiver, for utility 1.

(b) Free broadcast, saturating attention. Let the budget be nonbinding, let a receiver absorb one item freely, and let each further item cost it 1. Broadcasting S to everyone yields $|S| - n\max(0, |S|-1)$, maximized at $|S| = 1$ with value 1, while per-receiver allocation again yields n. □

Table 1: Existence of the two parameters that [19] requires *simultaneously*. Each phenomenon destroys exactly one.

	γ_1 (weak sub.)	γ_2 (weak super.)
Complementarity	**does not exist**	exists
Saturation	exists	**does not exist**

Case (b) is the more informative one. A reader may discount (a) as an artifact of charging per receiver; (b) removes that objection, because even with free broadcast the seed set is capped. The reason is worth stating plainly: *an additional broadcast item helps one receiver and harms the other $n-1$.*

Remark 1. The instance in case (a) contains no saturation and no complementarity: rel is modular and $\rho \equiv 0$, so the objective is monotone and submodular. The separation is therefore not a consequence of the degeneracies in §5. Repairing the objective's curvature does not rescue the seed-set formulation, because the loss comes from the choice of object, not from the objective's shape. Any guarantee proved against a seed-set optimum bounds distance from the wrong quantity.

5 Both toolchains degenerate

Monotonicity and submodularity fail. With rel modular and ρ increasing, an item whose relevance falls below its marginal saturation cost strictly lowers u_i, so U is not monotone. And two items individually inert but jointly decisive — $\mathrm{rel}(\emptyset) = \mathrm{rel}(\{a\}) = \mathrm{rel}(\{b\}) = 0$, $\mathrm{rel}(\{a, b\}) = 1$ — give increasing returns, so U is not submodular. Both are required by the greedy argument.

The nearest general framework has no parameters here. Shi and Lai [19] give guarantees for non-monotone, non-submodular maximization under a knapsack. Their general theorem requires the objective to be simultaneously γ_1-weak submodular and γ_2-weak supermodular:

$$F(U_2 \cup \{x\}) - F(U_2) \le \gamma_1 \left[F(U_1 \cup \{x\}) - F(U_1)\right], \quad (2)$$

$$\gamma_2 \left(F(U_2 \cup \{x\}) - F(U_2)\right) \ge F(U_1 \cup \{x\}) - F(U_1), \quad (3)$$

for $U_1 \subsetneq U_2$ and $x \notin U_2$. Our two phenomena destroy exactly one parameter each.

Under complementarity, take the instance above with $U_1 = \emptyset$, $U_2 = \{b\}$, $x = a$: (2) demands $1 \le \gamma_1 \cdot 0$, which no finite γ_1 satisfies. Under saturation, an item whose marginal is $+0.5$ at a lightly loaded receiver and -1.5 at a saturated one makes (3) demand $\gamma_2 \cdot (-1.5) \ge 0.5$, which no $\gamma_2 \ge 1$ satisfies.

Table 1 summarizes. The symmetry is the point: complementarity breaks weak *sub*modularity while saturation breaks weak *super*modularity, and since the theorem needs both at once it applies to neither phenomenon alone. This is a stronger statement than a loose bound — the parameters are undefined, so the guarantee is unavailable rather than weak.

6 Structure and algorithms

Proposition 1 says the per-receiver object is the right one. It does not say the resulting problem is intractable, and in one respect it is unusually well structured.

Proposition 2. *The problem is separable across receivers and coupled only by the shared budget. Consequently, writing $v_i(b) = \max\{u_i(S) : c(S) \le b\}$, an optimal allocation is obtained by solving*

$$\max \left\{ \textstyle\sum_i v_i(b_i) \ : \ \sum_i b_i \le B \right\}$$

and giving receiver i an optimal set at its level b_i. If each v_i is computed exactly, this is exact.

Proof. $U = \sum_i u_i(S_i)$ and the only constraint linking receivers is $\sum_i c(S_i) \le B$. Fix any feasible allocation and let $b_i = c(S_i)$. Then $u_i(S_i) \le v_i(b_i)$ and $\sum_i b_i \le B$, so its value is at most that of the displayed program. Conversely any feasible (b_i) with maximisers S_i is a feasible allocation of value $\sum_i v_i(b_i)$. □

Two consequences matter. First, the coupling between receivers is a single scalar: a budget price, recovered as the Lagrange multiplier of $\sum_i b_i \le B$. Second, and less obviously, the saturation penalty enters *only* through the chosen level b_i, so ρ_i may have any shape whatever — non-convex, discontinuous, a cliff. This matters because the measured degradation curves are S-shaped rather than convex, and a rule that reasons about ρ_i' requires convexity to argue that its admission bar rises as a receiver fills.

Two algorithm families follow, and we are explicit about what is new.

A density rule. The marginal of adding f to receiver i holding S is non-negative iff, to first order, $\Delta\mathrm{rel}_i(f \mid S)/c(f) \ge \rho_i'(\mathrm{conf}_i(S))$: send an item only when its relevance per token clears that receiver's current marginal degradation rate. An item failing the test both lowers u_i and consumes budget, so no optimal solution contains one — the restriction is a dominance property rather than an assumption. *This rule is not new.* Density-greedy with a positive-marginal filter is the distorted-greedy lineage [10], appears as Algorithm 1 of [19], and has been applied to token budgets for a single agent. We include it as a baseline for completeness.

Marginal allocation. The objective is separable across receivers and coupled only by the shared budget, so the problem factors: build each receiver's value-versus-budget curve $v_i(b)$ and solve $\max \sum_i v_i(b_i)$ subject to $\sum_i b_i \le B$ by dynamic programming. This equalizes marginal value per token *across* receivers, which neither a uniform per-receiver rule nor a per-receiver density rule can do — the former spends identically on every receiver, the latter has no view of what a token would buy elsewhere. The saturation penalty enters only through the chosen level, so it may take any shape, including the non-convex ones the measurements suggest. §7 reports that this does not, in fact, outperform a tuned uniform rule on our benchmark, and why.

7 Experiments

Setup. We use MuSiQue-Ans [21], where the structure we need is annotated: each instance supplies 20 candidate paragraphs with support labels, and sub-questions each carrying the index of the paragraph answering it. Receivers are sub-questions, the item pool is the paragraph set, and cost is tokens transmitted summed over receivers. Ground-truth relevance being labeled makes an oracle allocation computable.

Table 2: Paired comparisons, $n = 120$ receiver-instances, matched-budget grid. Δ is the first policy minus the second; token ratio > 1 means the first is cheaper. Starred $p < 0.05$.

Comparison	Δacc	p	tokens
per-receiver vs. seed set	+0.192	0.0004*	1.1×
oracle vs. broadcast	+0.050	0.146	21.4×
allocation vs. broadcast	−0.042	0.227	2.7×
uniform k=5 vs. broadcast	−0.083	0.013*	5.4×
density vs. uniform (matched)	+0.000	1.000	0.8×

Table 3: Accuracy and cost per policy, matched-budget grid, $n = 120$. Tokens are per receiver; the ratio is broadcast's tokens divided by the policy's. Absolute exact-match is suppressed in favour of deltas because the benchmark is partially memorised (§7); 5% of receiver-instances are answered correctly with no context at all.

Policy	tokens	ratio	Δacc vs. broadcast
broadcast (all items)	2258	1.0×	—
oracle (labelled)	99	21.4×	+0.050
per-receiver, uniform	418	5.4×	−0.083
density rule	502	4.4×	−0.083
marginal allocation	831	2.7×	−0.042
seed set (matched)	260	8.7×	−0.263
random (matched)	345	6.5×	−0.632
no context	0	—	−0.763

Two points of disclosure. MuSiQue is published as a single-agent task; splitting it across per-sub-question receivers is our framing. And most of its decompositions are sequential, so we restrict to the branching families in which two sub-questions carry no back-reference and are genuinely independent. Because the benchmark predates current models, some answers are memorized; an empty-allocation arm serves as a contamination control and we report policy *deltas* rather than absolute exact-match. Comparisons are paired — every policy answers the same items — using McNemar's exact test on discordant pairs.

The separation appears in the data. The strongest measured effect is the first row of Table 2: per-receiver allocation beats a seed-set baseline — items ranked by aggregate score and sent to everyone — by 19.2 accuracy points at matched cost. Proposition 1 predicts the gap should grow with the number of receivers, and it does: as we fuse instances to raise n from 2 to 8 to 16, seed-set delivery recall falls from 0.45 to 0.08 to 0.01 while per-receiver recall stays above 0.55.

Allocation is cheap. Allocating rather than broadcasting cuts tokens 2.7× with no measurable accuracy cost (row 3, $p = 0.227$). Pushing to 5.4× does cost 8 points, significantly (row 4), so the frontier is real. The oracle beats broadcast by $\sim$ 5 points at 21× fewer tokens, consistently across two models (+0.051 and +0.050) though significant on only one — we treat it as a consistent effect at borderline power rather than an established one.

Table 4: Proposition 1 predicts the seed-set gap grows with the number of receivers. Delivery recall as instances are fused to raise n. The seed set collapses; per-receiver allocation does not.

receivers n	2	8	16
seed set (matched cost)	0.45	0.08	0.01
per-receiver, uniform	0.67	0.56	0.56

Table 5: Replacing lexical relevance with a dense embedding scorer *harms* per-receiver retrieval but helps the seed set. The two signals fail in opposite directions: embeddings capture what an instance is about, lexical matching captures which receiver needs what.

Delivery recall	lexical	embedding
per-receiver, top-1	**0.43**	0.25
per-receiver, top-5	**0.79**	0.61
seed set, k=3	0.45	**0.85**

A negative result on algorithm design. Across four value-curve designs, marginal allocation does not beat a tuned uniform per-receiver rule at matched budget; nor does the density rule (row 5, exactly zero at $p = 1.000$ while spending 25% more). We also found that replacing lexical relevance with a dense embedding scorer makes per-receiver retrieval *worse* (0.43 $\rightarrow$ 0.25 top-1 recall), because the sub-questions are terse relation templates whose entity identity a dense encoder discards — while the same change nearly doubles seed-set recall, which needs only instance-level topicality.

Three independent observations thus point the same way: the binding constraint is relevance *estimation*, not allocation *optimization*. The oracle fixes the scale of it, reaching full delivery on 208 tokens where the best implementable policy reaches 62% on 929. No allocation algorithm closes that gap, because the information required is not present in the scores.

8 Discussion

The results divide cleanly. The formulation-level claims are strong: the seed-set object is wrong by a factor $\Theta(n)$ even when the objective is well behaved, and both the classical toolchain and its nearest general successor are unavailable here. The algorithm-level claims are deliberately weak: we do not have a method that beats a tuned uniform per-receiver rule, and we report that rather than selecting a favorable budget point.

We take the negative result to be informative rather than merely disappointing. Marginal allocation can only help where per-receiver value curves are distinguishable from label-free signals, and on this benchmark they are not — every receiver draws from a common pool with a similar score distribution, so an equal split is already near optimal. Heterogeneity in what receivers *need* does not surface as heterogeneity in what can be *estimated* about them. Settings where it does — receivers with genuinely different and predictable appetites — are where the separation's practical value should be sought.

Limitations. The multi-receiver framing of a single-agent benchmark is ours. Our positive result is a separation, not an approximation guarantee: we show what fails without supplying a ratio for the multi-receiver problem, which we regard as the main open question. The saturation penalty is calibrated from published degradation curves whose magnitude is model- and task-dependent, and at least one study reports large models resisting non-confusable distractors almost entirely. Finally, the agent graph is treated as given; in deployed systems an orchestrator builds it in response to the task, so transmissions reshape the topology they traverse, and we have no formal treatment of that feedback.

9 Conclusion

Most inter-agent context traffic can be removed for free. A 2.7× reduction costs nothing measurable, and an oracle reaches 21× while improving accuracy, so the ceiling is roughly an order of magnitude rather than a few percent. That is the practical headline, and it holds on a public benchmark under paired testing.

Reaching the ceiling is a different matter, and the obstruction is not where we expected. Casting the problem as influence maximization is the natural move and the wrong one. The obstruction is not that agents degrade under irrelevant context, though they do, nor that information items interact, though they do; it is that the seed-set object — one set serving a network — is off by a factor $\Theta(n)$ from the per-receiver object the problem actually poses, on instances where the objective is as well behaved as one could ask. Two toolchains that would otherwise apply are unavailable, the second because its structural parameters do not exist rather than because its bound is loose.

But the route to the remaining order of magnitude does not run through better allocation algorithms — which is what we expected, and not what we found. It runs through better estimates of which receiver needs what. For a practitioner the implication is concrete: switch from broadcasting to per-receiver allocation, which is nearly free and worth 19 points over the globally-scored alternative, then spend the next increment of effort on retrieval rather than on the allocation rule.

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